4.8.1 · 4.8.2 · 4.8.3 Transformer voltage, power and efficiency

4.8.1 Transformer voltage and turns ratio

In an ideal transformer, primary and secondary windings link the same changing flux, so their voltage ratio equals their turns ratio. More secondary turns give step-up action and fewer give step-down action. Steady d.c. cannot maintain changing flux and does not transform normally. Compare either both r.m.s. voltages or corresponding instantaneous values.

$$\frac{V_p}{V_s} = \frac{N_p}{N_s}$$

Ideal transformer: primary/secondary voltages V_p, V_s (V), and turns N_p, N_s. Alternating flux; use matching RMS or instantaneous values.

4.8.2 · 4.8.3 Transformer power and efficiency

An ideal transformer has equal input and output powers, so stepping up voltage reduces the corresponding available current. Real transformers have winding and core losses; output power equals input power times efficiency. Convert percentage efficiency to a fraction before calculation. For resistive a.c. loads, use r.m.s. voltage and current.

$$V_p I_p = V_s I_s$$

$$\eta V_p I_p = V_s I_s$$

Ideal transformer with no power loss and resistive load; p: primary; s: secondary; voltage in V and current in A.

Efficiency η = output/input power, written as a fraction from 0 to 1. Divide a percentage by 100 before substitution. Resistive load; RMS AC values.

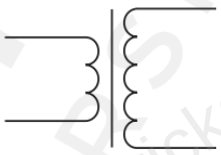
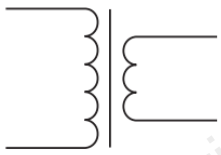
	Step-up transformer	Step-down transformer
Voltage	$V_\mathrm{p} < V_\mathrm{s}$	$V_\mathrm{p} > V_\mathrm{s}$
Number of turns	$N_\mathrm{p} < N_\mathrm{s}$	$N_\mathrm{p} > N_\mathrm{s}$
Symbol		

Table 8.1a *Step-up transformer and step-down transformer.*

Step-up: more secondary turns and a higher secondary voltage. Step-down: fewer secondary turns and a lower voltage; ideal power remains conserved.

4.8.4 High-voltage transmission

For the same transmitted power, higher voltage allows lower current. Cable heating loss depends on current squared, so reducing current reduces loss. Distinguish supply voltage, the voltage drop along the cable and the cable's dissipated power. Do not square the whole transmission voltage and divide by cable resistance to calculate its loss.

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